

Why is countermovement jump height greater than squat jump height?

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ABSTRACT

BOBBERT, M. F., K. G. M. GERRITSEN, M. C. A. LITJENS, and A. J. VAN SOEST. Why is countermovement jump height greater than squat jump height? *Med. Sci. Sports Exerc.*, Vol. 28, No. 11, pp. 1402–1412, 1996. In the literature, it is well established that subjects are able to jump higher in a countermovement jump (CMJ) than in a squat jump (SJ). The purpose of this study was to estimate the relative contribution of the time available for force development and the storage and reutilization of elastic energy to the enhancement of performance in CMJ compared with SJ. Six male volleyball players performed CMJ and SJ. Kinematics, kinetics, and muscle electrical activity (EMG) from six muscles of the lower extremity were monitored. It was found that even when the body position at the start of push-off was the same in SJ as in CMJ, jump height was on average 3.4 cm greater in CMJ. The possibility that nonoptimal coordination in SJ explained the difference in jump height was ruled out: there were no signs of movement disintegration in SJ, and toe-off position was the same in SJ as in CMJ. The greater jump height in CMJ was attributed to the fact that the countermovement allowed the subjects to attain greater joint moments at the start of push-off. As a consequence, joint moments were greater over the first part of the range of joint extension in CMJ, so that more work could be produced than in SJ. To explain this finding, measured and manipulated kinematics and electromyographic activity were used as input for a model of the musculoskeletal system. According to simulation results, storage and reutilization of elastic energy could be ruled out as explanation for the enhancement of performance in CMJ over that in SJ. The crucial contribution of the countermovement seemed to be that it allowed the muscles to build up a high level of active state (fraction of attached cross-bridges) and force before the start of shortening, so that they were able to produce more work over the first part of their shortening distance.

PRE-STRETCH, FORCE DEVELOPMENT, ELASTIC ENERGY,
MUSCLE MODELING

When performing motor tasks such as throwing and jumping, human beings typically start with a so-called countermovement, a movement in a direction opposite to the goal direction. There is ample evidence that task performance is improved by such a countermovement. For instance, it has been shown that subjects achieve a greater jump height in a so-called countermovement jump (CMJ), where they start from an

erect position and make a downward movement before starting to push-off, than in a so-called squat jump (SJ), where they are instructed to start from a semi-squatted position and make no countermovement. This is true even if body configuration at the start of push-off is the same (1). Apparently, the subjects are able to produce more work and/or use the work produced more effectively in CMJ than in SJ. To explain the difference in performance between CMJ and SJ, several possible explanations may be offered.

The first possible explanation is that subjects are simply not used to making SJ, and as a consequence are unable to properly control this type of jump. If control, also referred to as coordination, is not optimal, actual jump height will be less than the maximum achievable height determined by the properties of the musculoskeletal system. The possibility that optimal control for SJ is different from optimal control for CMJ should be considered, because differences in initial conditions between CMJ and SJ might require adaptation of control. Nonoptimal control is likely to affect the movement pattern and cause the fraction of muscle work transformed to effective energy (energy contributing to jump height) to be submaximal.

A second possible explanation is that in SJ the muscles are unable to achieve a high level of force prior to the start of concentric contraction. In a maximal voluntary muscle contraction, it takes time before the muscle force has reached a maximum value. This is due partly to the finite rate of increase of muscle stimulation by the central nervous system (stimulation dynamics), partly to the time constants of the stimulation-active state coupling (excitation dynamics), and partly to the interaction between contractile elements and series elastic elements (contraction dynamics). If the concentric active state starts as soon as the force begins to rise, part of the shortening distance of the muscle-tendon complexes is traveled at a submaximal force, and thus the work produced is submaximal (11,24). This undesirable effect can be avoided by allowing the muscle to build up a maximum active state before the start of the concentric contraction, either

in an isometric contraction (4,6) or during a counter-movement, as in CMJ.

A third possible explanation concerns the storage and reutilization of elastic energy (2,25). The idea is that during the countermovement in CMJ active muscles are pre-stretched and absorb energy, part of which is temporarily stored in series elastic elements and later reutilized in the phase where the muscles act concentrically. It is argued that this helps to increase the work produced in CMJ over that produced in SJ (2,25).

A fourth possible explanation is that the muscle stretch that occurs during the countermovement in CMJ triggers spinal reflexes (13) as well as longer-latency responses (26), that help to increase muscle stimulation during the concentric phase to a level surpassing that achieved in SJ, where no pre-stretch occurs. At this higher stimulation level, the muscles may produce a larger force, and thus more work during the concentric phase.

A final possible explanation is that the pre-stretch of active muscle, which occurs during the countermovement in CMJ, alters the properties of the contractile machinery. It is well-documented that the force produced by artificially stimulated isolated muscles may be enhanced by pre-stretch (8–10,17–19) and the same has been found for tetanized single muscle fibers (14,15). This enhancement, also called potentiation (7), has been shown to increase with the speed of pre-stretch (14,15) and to decrease with the amount of time elapsed after the pre-stretch (10,14,15). Again, enhancement of force in CMJ could help to increase the work produced in CMJ over that produced in SJ.

The purpose of this study was to investigate the role of the time available for force development and the storage and reutilization of elastic energy in enhancing countermovement jump performance over squat jump performance. For this purpose, kinematics, kinetics, and muscle electrical activity were monitored during maximum height countermovement and squat jumps of human subjects. The squat jumps were performed from different starting positions, to study the effect of this variable on jump height. Furthermore, kinematics and electromyographic activity were used as input for a model of the musculoskeletal system that calculated internal states and forces of individual muscle-tendon complexes. The model allowed for an investigation of the effects of the different stimulation histories in CMJ and SJ on joint moments and joint work.

METHODS USED IN EXPERIMENTAL STUDY

Subjects and Outline of Experimental Setup

Six male volleyball players (Dutch Division of Honour) participated in this study. The reason for selecting these subjects was that they were used to performing both CMJ and SJ in training practice and competition. In-

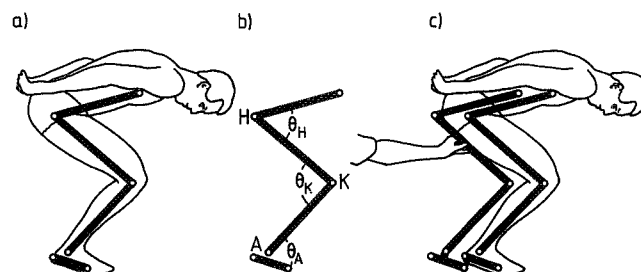


Figure 1—(a) The body was represented by four rigid segments with the help of an “exoskeleton.” (b) Definition of joint angles (θ) at hip (H), knee (K), and ankle (A). (c) To adjust the starting position in squat jumps, the subject was asked to reproduce the configuration of a second “exoskeleton,” which was held next to the body and used as a template.

formed consent was obtained from all subjects in accordance with the policy statement of the American College of Sports Medicine. Characteristics of the group of subjects (mean $\pm$ standard deviation) were: age 25 ± 4 yr, height 1.93 ± 0.08 m, body mass 79.6 ± 5.8 kg.

For the study of differences in jump height and mechanical output between CMJ and SJ each subject performed, while holding the arms behind the back (Fig. 1a), maximum height jumps of four different types: 1) CMJ: countermovement jumps according to the subject's own preferred style; 2) SJC: squat jumps starting from a posture that was identical to the posture during CMJ at the start upward movement of the mass center of the body; 3) SJP: squat jumps starting from the posture preferred by the subject; 4) SJD: squat jumps starting from a posture that was as low as possible.

During the squat jumps, the subjects were instructed to make no countermovement. After a number of practice trials, they were able to perform the jumps as required. For the actual experiment, each subject performed jumps of the different types in random order, until three successful jumps of each type had been made. During each jump, sagittal-plane position data of anatomical landmarks were monitored, ground reaction forces were measured, and electromyograms were recorded from six muscles of the lower extremity. Jump height, defined as the difference between the height of the mass center of the body at the apex of the jump and the height of this mass center when the subject was standing upright with heels on the ground, was calculated from the position data. The highest jump of each type was selected for further analysis. Net joint moments and work were obtained by performing an inverse-dynamics analysis, combining kinematic information and ground reaction forces. Details on methods and procedures are provided below.

Collection and Processing of Position Data and Adjustment of Initial Posture in SJ

For definition of the body position and determination of body segment kinematics, an “exoskeleton” was used,

consisting of four inextensible rods, three of which were interconnected by hinge joints (Fig. 1B). The rods had retroreflective spheres of 2 cm diameter on their free ends and at the height of the joints. By adjusting the positions of the joints on the rods, the effective lengths of the rods were adjusted to each subject. The adjustment was such that in the sagittal plane the upper rod represented the orientation of the trunk, the most proximal hinge joint the location of the estimated flexion/extension axis of the hip joint (at the height of the greater trochanter), the second rod the orientation of the upper legs, the second hinge joint the estimated flexion/extension axis of the knee joints (at the height of the lateral femoral epicondyle), the third rod the orientation of the lower legs, the distal end of the third rod the estimated dorsiflexion/plantarflexion axis at the ankle joint (0.5 cm anterior to the tip of the lateral malleolus), and the fourth rod the orientation of the feet. The rods were affixed firmly to the body by means of athletic tape and elastic bandages. Placement of the exoskeleton and definition of joint angles are illustrated in Figures 1a and 1b, respectively.

The locations of the spheres in space were monitored using four electronically shuttered cameras (NAC 60/200 MOSTV) connected to a VICON high-speed video analysis system (Oxford Metrics Ltd., Oxford, U. K.) operating at 200 Hz, and three-dimensional coordinates were reconstructed from individual camera views using AMASS software (Oxford Metrics Ltd., Oxford, U. K.). Only sagittal plane projections were used in this study. The locations of the mass centers of upper legs, lower legs, and feet were estimated from the landmark coordinates, in combination with results of cadaver measurements presented in the literature (12). To determine the location of the mass center of the upper body relative to the two markers on this segment, the subject was asked to assume two equilibrium postures on tiptoes, one with the hip joints fully extended, one with the hip joints flexed. This yielded two equations for the fore-aft coordinate of the mass center of the body, which in the case of equilibrium equals the fore-aft coordinate of the center of pressure on the force-plate. The two equations were solved for the two unknown coordinates of the mass center of the upper body relative to the markers on this segment. With this information, the position of the mass center of the body was calculated in all other body positions found during jumping.

Before the start of the actual experiments, the subjects performed three maximal CMJ. For each of these, the vertical coordinate of the mass center of the body (MCB) was calculated as a function of time, and the highest jump of the three was selected. For this highest jump, the frame where MCB attained its minimum height was identified, and the body configuration on this frame was copied onto a second "exoskeleton," which could be held next to the subject as a template for adjustment of the initial posture in SJC (Fig. 1c). Henceforth, the phase preceding the

instant where MCB attained its lowest position will be referred to as countermovement phase, and the phase between this instant and toe-off as push-off phase.

For kinetic analysis (see below), the time histories of marker positions were smoothed using a bidirectional low-pass Butterworth filter with a cutoff frequency of 16 Hz. To obtain linear velocities and accelerations, the smoothed position time histories were differentiated numerically with respect to time using a direct 5-point derivative routine. Angles of body segments with respect to the horizontal were calculated from the smoothed marker position time histories, and differentiated to obtain angular velocities and accelerations.

Collection and Processing of Ground Reaction Force Data

The three orthogonal components and the point of application of the ground reaction force vector were determined using a force plate (KISTLER type 9281B, Kistler Instrumente AG, Winterthur, Switzerland), connected to an electronic amplifier unit (KISTLER type 9861A, Kistler Instrumente AG, Winterthur, Switzerland). The eight output signals of this unit were sampled simultaneously with the position data at 200 Hz using the analog-to-digital convertor of the VICON system, and used to calculate the vertical and fore-aft components of the ground reaction force vector and the fore-aft coordinate of its point of application.

Kinetic Analysis

Net intersegmental forces and moments were calculated following a standard inverse-dynamics approach (16) using the measured components and point of application of the ground reaction force vector, locations of joint axes and segmental mass centers obtained from the position data, linear and angular accelerations of segments derived from position data, segmental masses estimated using data reported by Clauser et al. (12), and transverse segmental moments of inertia obtained using the estimates of the previous segmental properties in combination with values reported by Plagenhoeff et al. (28) on segmental radii of gyration. Hip extension, knee extension, and plantarflexion moments were defined as positive.

Collection and Processing of Electromyograms

Electromyograms were recorded using pairs of silver-chloride EKG surface electrodes (Sentry Medical Products, Irvine, CA, center-to-center distance 4 cm) from m. biceps femoris, m. gluteus maximus, m. rectus femoris, m. vastus medialis, m. gastrocnemius, and m. soleus. The positioning of the electrodes over the muscles is described elsewhere (20). Movement of cables, and therefore movement artifact, was minimized by wrapping the

cables to the skin with the bandages used to fasten the exoskeleton. EMG signals were pre-amplified, transmitted, and further amplified (BIOMES 80, Glonner Electronics GmbH, Munich, Germany), high-pass filtered at 7 Hz to reduce the amplitude of possible movement artifacts, full-wave rectified, smoothed using an analog 20 Hz 3rd order low-pass filter, and sampled at 200 Hz using the analog-to-digital convertor of the VICON system. To further reduce the variability of each sampled electromyographic signal in time, it was smoothed off-line using a bidirectional digital low-pass Butterworth filter with a 7-Hz cutoff frequency, to yield SREMG (Smoothed Rectified EMG). Details and motivation of this processing technique are presented elsewhere (4). SREMG values were subsequently normalized to NSREMG by expressing them as fraction of the maximum value attained during the four selected jumps (one value for each subject).

Processing of Data and Statistical Analysis

For comparison of variables among the four different types of jumps, analysis of variance for single factor experiments with repeated measures on the same elements was used in combination with a Newman-Keuls *post-hoc* test (32). For comparison of differences between CMJ and SJC, a Student's *t*-test for paired comparisons was used. The level of significance for all tests was 0.05.

METHODS USED IN SIMULATION STUDY

Outline of the Simulation Study

To explain differences in jump height and mechanical output at joints between CMJ and SJC in the experimental study, individual angle-time and NSREMG-time curves were used as input for a model of the musculoskeletal system, schematically shown in Figure 2a. The output of the model comprised internal states and forces of individual muscle-tendon complexes, as well as net joint moments. First, for each subject, scaling factors for NSREMG as well as maximal isometric forces of the muscles were obtained by optimization, using as criterion the sum of squared differences between calculated and measured moment time histories. Subsequently, to study the effect of the dynamics of force development on work produced, NSREMG-time curves of the phase preceding the start of push-off in CMJ were replaced by those of the corresponding phase in SJC. Finally, to study the effect of storage and reutilization on work performed, a simplified version of the model (Fig. 2b) was used to simulate hypothetical single-joint contractions. Details of the simulation study are provided below.

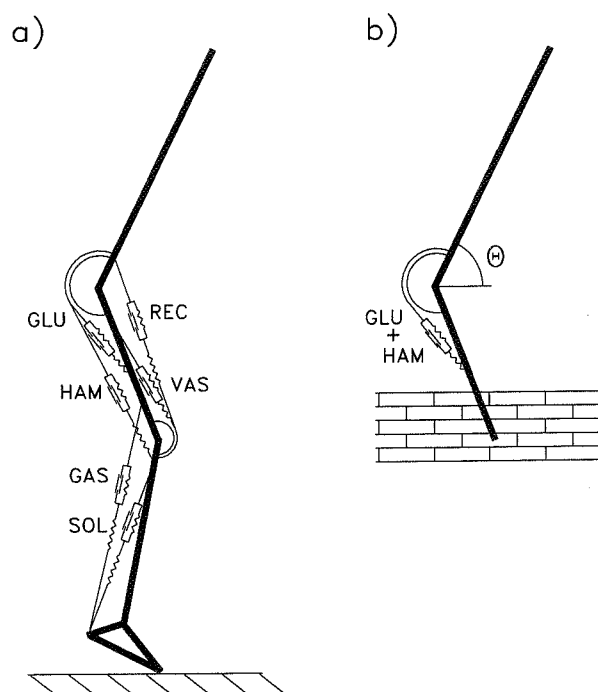


Figure 2—(a) Schematic drawing of the model of the musculoskeletal system used in simulation of countermovement and squat jumps. It consists of four rigid segments (feet, lower legs, upper legs, and head-arms-trunk) and six muscle-tendon complexes of the lower extremity (GLUTEi, HAMstrings, VASTi, RECTus femoris, SOLEus, and GASTrocnemius), all represented by Hill-type muscle models. (b) Simplified version of model, used for investigation of the role of elastic energy. The only segment that is free to rotate is head-arms-trunk, and the only actuators are the hip extensor muscles.

Model of Musculoskeletal System

The model of the musculoskeletal system, used to calculate internal states and forces of individual muscle-tendon complexes, was described elsewhere (29). It consists of four rigid segments representing feet, lower legs, upper legs, and head-arms-trunk, which are interconnected by hinge joints representing hip, knee, and ankle joints. Six major muscle-tendon complexes contributing to extension of the lower extremity are embedded in this skeletal model: GLUTEi, HAMstrings, VASTi, RECTus femoris, SOLEus, and GASTrocnemius. A Hill-type muscle model was used to represent each of these six complexes. It consists of a contractile element, a series elastic element and a parallel elastic element, and is described in full detail elsewhere (29). Behavior of the elastic elements is defined by a nonlinear force-length relationship. Behavior of the contractile element is more complex: contraction velocity depends on active state, contractile element length, and force, with force being directly related to the length of the series elastic element. This length can be calculated at any instant from joint angles and contractile element length (selected as state variables) since muscle-tendon complex length is directly related to position of the skeleton. As proposed by Hatze (22,23), the relationship between active state (represent-

ing the fraction of cross-bridges attached) and muscle stimulation (STIM) was modeled as a first-order process. That is, active state was related according to a sigmoid curve to the free calcium concentration γ , which in turn was related to STIM by first-order dynamics: $\dot{\gamma} = m \cdot (c \cdot \text{STIM}(t) - \gamma)$, where m and c are constants and STIM, ranging between 0 and 1, is a one-dimensional representation of the effects of recruitment and firing frequency of α -motoneurons.

Determination of Parameter Values of the Model

The interaction between muscles and skeleton is governed by two relationships: the relationship between joint angles and lengths of muscle-tendon complexes, and that between joint angles and moment arms of these complexes. For the first relationship, data were used from cadaver studies in which muscles were cut and the gap width between the corresponding ends was measured over a range of joint angles (21,31). For the second relationship, the first derivative of these curves with respect to joint angle was used. Moment arm values of the model can be found elsewhere (30).

In determining parameter values for the behavior of muscle-tendon complexes, several conceptual steps were made (5). First, a muscle-tendon complex was regarded as a collection of identical units, each composed of a muscle fiber made up of identical sarcomeres in series, and a "tendon fiber" bridging the gap between muscle fiber length and origin-to-insertion distance. Second, it was assumed that the muscle fibers were responsible for the properties of the contractile and parallel elastic elements, and the "tendon fibers" for those of the series elastic element. Given the properties of each sarcomere and the stiffness properties of tendinous tissue (29), these assumptions essentially reduce the problem of finding parameter values for the behavior of the muscle-tendon complex to that of determining the number of sarcomeres in series in each muscle fiber, the number of sarcomeres in parallel in the whole muscle, and the origin-to-insertion distance at which the muscle-tendon complex attains its optimal length (i.e., the length where it produces its maximum isometric force). The number of sarcomeres in series was actually counted in muscle fibers obtained from cadavers (Huijing, personal communication), and this allowed for the determination of the relationship between muscle fiber length and normalized force. Optimal muscle fiber lengths used in the model, i.e., the product of optimal sarcomere length and the number of sarcomeres in series, can be found elsewhere (30).

Values for absolute muscle forces and tendon fiber slack lengths were estimated as follows. Physiological cross-sectional areas of muscles, defined as muscle volume divided by muscle fiber optimal length, were determined in human cadavers (Huijing, personal communication). The ratio of maximal isometric forces of the

muscles crossing a joint was set equal to the ratio of physiological cross-sectional areas. The following ratios were derived for the muscle groups of the model shown in Figure 2a: GLU:HAM = 5:4, VAS:REC = 3:1, and SOL:GAS = 2:1. Using an optimization procedure, tendon fiber slack lengths of the muscles crossing a joint were adjusted in such a way that the best fit was obtained between the normalized maximum isometric moment-angle relationship of the model and that measured in maximum isometric contractions of subjects on a dynamometer (27). Finally, a total maximal isometric force for the hip extensor $F_{\max, \text{HE}}$, knee extensors $F_{\max, \text{KE}}$, and plantar flexors $F_{\max, \text{PF}}$ of each subject was found by means of optimization, as will be explained below. These total maximal isometric forces were distributed among the individual muscles using the ratios mentioned above.

Individualization of the Model

Independent input of the model were individual time histories of joint angles (and therewith muscle-tendon complex lengths) and STIM of the muscles. Joint angle time histories were obtained directly from the experimental study, STIM time histories were assumed to correspond to the measured NSREMG time histories, scaled by a fraction representing the stimulation fraction of each individual muscle. Thus, the differential equation for the free calcium concentration became: $\dot{\gamma}_i = m \cdot (f_i \cdot \text{NSREMG}_i(t) - \gamma_i)$.

Since the purpose of this study was not to predict the mechanical output about joints during CMJ and SJC, but to try and explain differences in mechanical output between these jumps, it was considered permissible to tune the model slightly to the individual experimental data. To this end, the f_i -values together with $F_{\max, \text{HE}}$, $F_{\max, \text{KE}}$, and $F_{\max, \text{PF}}$ were optimized for each subject, using as criterion the time-integral of squared differences between calculated and measured net joint moments, summed over CMJ and SJC. In the optimization procedure, a nonlinear parameter transformation was used to ensure that the f_i -values remained between 0 and 1.

RESULTS AND DISCUSSION

We shall first present and discuss experimentally observed differences in performance between countermovement and squat jumps, and the effects of starting position. Subsequently, a comparison will be made between experimental and simulated joint moment time histories. After this model evaluation, we shall present and discuss the results of simulations, which were carried out to investigate the role of the time available to develop muscle force and the role of storage and reutilization of elastic energy. Finally, we shall discuss the potential roles of reflexes and potentiation, which were not specifically investigated in this study.

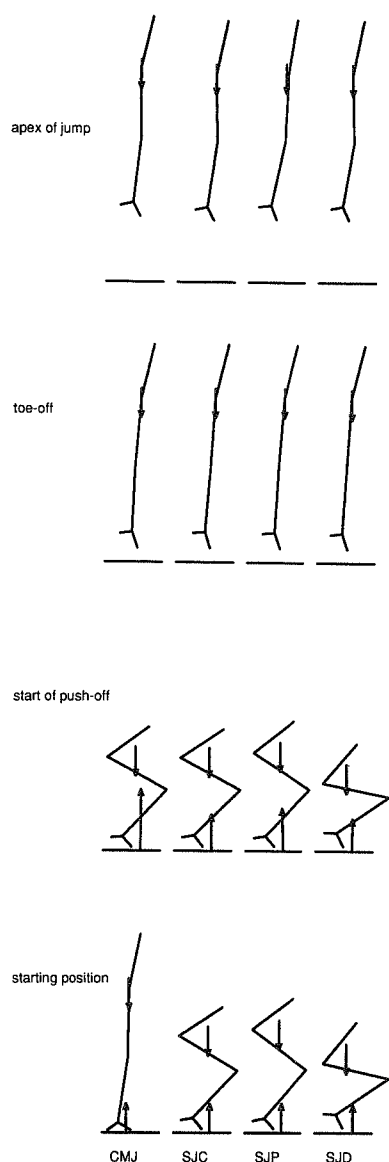


Figure 3—Average stick diagrams of the group of subjects ($N = 6$) for the four types of jumps performed in this study. Stick diagrams are shown for starting position, start of push-off, toe-off, and apex of the jump. In each diagram, the vector of the force of gravity is shown with its origin in the location of the center of mass, and the ground reaction force vector is shown with its origin at the center of pressure on the force platform. CMJ: countermovement jump; SJC: squat jump starting from a posture that was identical to that at the start of push-off in CMJ; SJP: squat jump starting from the posture preferred by the subject; SJD: squat jump starting from a posture that was as low as possible. Note that the foremost foot marker was on the Vth metatarsophalangeal joint, which explains the “premature” upward movement of this marker.

Experimental Findings

The differences in performance between CMJ and SJ are summarized in Figure 3 and Tables 1 and 2. Figure 3 shows average stick diagrams for the four types of jumps performed in this study at (a) starting position, (b) start of push-off, (c) toe-off, and (d) apex of the jump. In each diagram, the vector of the force of gravity is shown with

its origin in the location of MCB, and the ground reaction force vector is shown with its origin at the center of pressure on the force platform. The diagrams illustrate a number of key results, which are quantitatively supported by values for selected variables related to the movement of the center of mass (Table 1) and variables for the individual joints (Table 2). First of all, they show that the starting body configuration in SJC was successfully controlled: neither the height of MCB (Table 1) nor joint angles (Table 2) in this position or the position at the start of push-off in SJC were significantly different from those in the position at the start of push-off in CMJ ($P > 0.05$). Second, at toe-off, body position was the same in all jumps (Table 2). However, as can be seen in Table 1, jump height was more than 2.5 cm greater in CMJ than in all other jumps ($P < 0.05$) as the result of a greater vertical velocity at toe-off ($P < 0.05$). Third, a major difference between CMJ and the other jumps occurred in the magnitude of the ground reaction force at the start of push-off: in CMJ, it was much greater than in the other jumps (Table 1). From Table 2 it can be seen that this greater reaction force is due to greater hip extension moments, knee extension moments, and plantarflexion moments at the start of push-off ($P < 0.05$).

These findings first of all confirm the conclusion of previous studies that subjects are able to jump less high in SJ than in CMJ. They also show that when the starting position is not controlled (SJP), subjects lower their mass center less in SJ than in CMJ (Table 1, $P < 0.05$). Consequently, the distance over which they can produce force is now smaller, but this does not explain the difference in jump height. After all, in SJD the subjects made a much deeper squat than in CMJ, but still jumped on average 3.2 cm less high than in CMJ (Table 1, $P < 0.05$). The greater jump height in CMJ than in the other jumps seems to be due primarily to the fact that the countermovement allowed the subjects to attain greater joint moments at the start of push-off. As a consequence, joint moments were also greater over the first part of the range of joint extension in CMJ, so that more work could be produced than in the squat jumps. This is illustrated in Figure 5a (bottom), which shows average joint moment-angle curves for the group of subjects. The area under these curves represents mechanical work. At the hip joints, more work is produced in CMJ (on average 268 J in CMJ versus 242 J in SJC, $P < 0.05$).

The experimental results confirm that a greater jump height can be achieved in CMJ than in SJ, and indicate that the greater achievement in CMJ is attributed to the fact that more work can be produced after the countermovement. The challenge now is to identify the mechanism(s) responsible for the enhancement of work output in CMJ. In the experimental results there is no indication that coordination plays a role. There are no signs of movement disintegration in any of the SJ, and in all jumps convergence to the same toe-off position occurred

TABLE 1. Mean values ($\pm$ SD) for selected variables related to movement of the center of mass of the body in the group of subjects ($N = 6$) during different types of jumps.

	CMJ	SJC	SJP	SJD
$Z_{MCB,spo} - Z_{MCB,ref}$ [m]	-0.350 ± 0.034	-0.356 ± 0.023	-0.319 ± 0.018	-0.469 ± 0.077
$Z_{MCB,to} - Z_{MCB,ref}$ [m]	0.119 ± 0.015	0.119 ± 0.014	0.119 ± 0.014	0.113 ± 0.024
$\dot{Z}_{MCB,to}$ [m/s]	$2.78^* \pm 0.10$	2.66 ± 0.15	2.66 ± 0.16	2.65 ± 0.17
$Z_{MCB,max} - Z_{MCB,ref}$ [m]	$0.481^* \pm 0.036$	0.447 ± 0.042	0.455 ± 0.049	0.449 ± 0.048
$t_{to} - t_{spo}$ [s]	$0.33^* \pm 0.03$	0.42 ± 0.06	0.36 ± 0.06	0.54 ± 0.06
$F_{z,spo}$ [N]	$1708^* \pm 336$	1006 ± 218	1187 ± 335	905 ± 142

CMJ: countermovement jump.

SJC: squat jump starting from a posture that was identical to that at the start of push-off in CMJ.

SJP: squat jump starting from the posture preferred by the subject.

SJD: squat jump starting from a posture that was as low as possible.

 $Z_{MCB,spo} - Z_{MCB,ref}$: height of the mass center of the body (MCB) at the start of push-off relative to reference height, i.e., height of MCB in upright standing. $Z_{MCB,to} - Z_{MCB,ref}$: height of MCB at toe-off relative to reference height. $\dot{Z}_{MCB,to}$: vertical velocity of MCB at toe-off. $Z_{MCB,max} - Z_{MCB,ref}$: jump height, i.e., maximum height reached by MCB relative to reference height. $t_{to} - t_{spo}$: duration of the push-off phase. $F_{z,spo}$: vertical component of the ground reaction force at the start of push-off.Asterisks indicate that the value in CMJ differs from that in the other jumps ($P < 0.05$).TABLE 2. Mean values ($\pm$ SD) for selected variables related to joint rotations in the group of subjects ($N = 6$) during different types of jumps.

	Ankle Joints		Knee Joints		Hip Joints	
	CMJ	SJC	CMJ	SJC	CMJ	SJC
θ_{spo} [rad]	1.26 ± 0.22	1.20 ± 0.16	1.31 ± 0.09	1.27 ± 0.18	1.12 ± 0.23	1.08 ± 0.17
θ_{to} [rad]	2.18 ± 0.11	2.19 ± 0.12	3.10 ± 0.08	3.03 ± 0.12	2.94 ± 0.090	2.90 ± 0.12
M_{spo} [Nm]	$220^* \pm 60$	127 ± 49	$289^* \pm 76$	179 ± 46	$330^* \pm 96$	177 ± 78

CMJ: countermovement jump.

SJC: squat jump starting from a posture that was identical to that at the start of push-off in CMJ.

 θ_{spo} : joint angle at the start of push-off. θ_{to} : joint angle at toe-off. M_{spo} : joint moment at the start of push-off.Asterisks indicate that the value in CMJ differs from that in SJC ($P < 0.05$).

(Fig. 3, Table 2). To be able to investigate the possible role of the time available to develop muscle force and storage and reutilization of elastic energy, we needed the help of the simulation model.

Comparison of Simulated and Experimental Joint Moment Histories

Figure 4 shows for one subject time histories of joint angles (a), NSREMG (b), and net joint moments (c), as well as time histories of net joint moments calculated by the model for CMJ and SJC on the basis of joint angles and NSREMG time histories (d). The joint moment histories calculated by the model (Fig. 4d) were the best approximation to the joint moment histories produced by the subject (Fig. 4c) that could be achieved by optimizing f_i -values together with $F_{max,HE}$, $F_{max,KE}$, and $F_{max,PF}$, as explained in the Methods section. Table 3 presents the average values of the parameters found in the optimization and Figure 5b shows the average moment-time curves (top panel) and moment-angle curves (bottom panel) calculated by the model.

As can be seen in Table 3, the average f_i -values ranged from 0.46 to 0.67. It is important to realize that lowering f_i from 1.0 to 0.46 primarily affects the rate of change of active state in response to fluctuations in NSREMG, and has only little effect on the maximum level of active state that can be reached. This is due to the fact that the sigmoid relationship between γ and active state saturates at relatively low values of γ (22). For instance, in a

steady-state, a STIM of 0.46 will lead to a γ -level that corresponds to an active state of 97% of maximum. Despite the optimization of f_i and maximal isometric forces, the moment histories calculated by the model do not perfectly match the experimental ones (compare Fig. 4d with Fig. 4c, and Fig. 5b with Fig. 5a). This is, of course, not surprising. The model of the musculoskeletal system is a gross simplification of reality, and to arrive at values for most of the parameters we can do no better than make educated guesses. Moreover, NSREMG(t), which was used as a measure of STIM(t) in the model and dominates the calculated net joint moment time histories, is in reality an output signal of the muscle that is related in a rather fuzzy way to the true stimulation input. Despite these limitations, however, the model does reproduce the key difference between CMJ and SJC: just as in the experimental results (Fig. 5a), the SJC hip and knee moments lag behind their CMJ companions over the first part of the push-off, which causes the work produced over the first part of the range of joint extension to be lower in SJC than in CMJ. We therefore felt that the model was suitable to investigate the role of the amount of time available for force development and the role of storage and reutilization of elastic energy.

The Role of Time Available to Develop Muscle Force

The first question tackled with help of the simulation model was whether the difference in joint moments at the

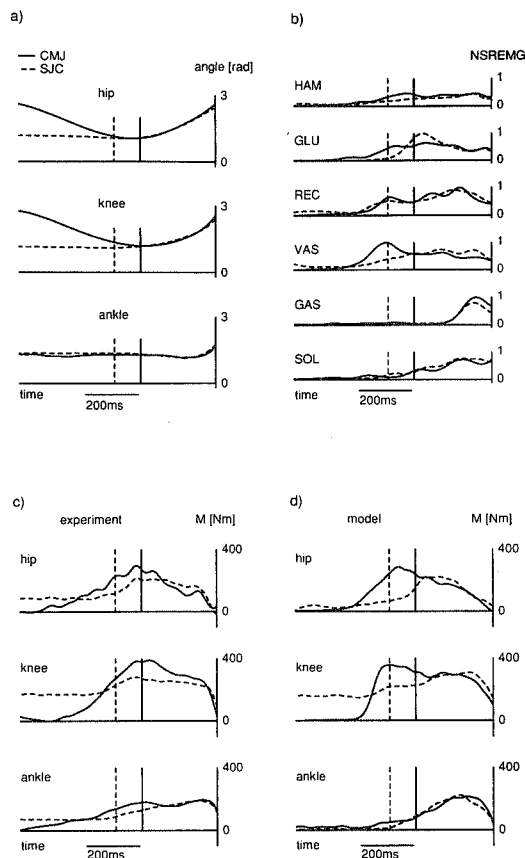


Figure 4—Typical individual time histories of (a) joint angles, (b) NSREMG (normalized smoothed rectified EMG), and (c) net joint moments found in the experiment, as well as (d) time histories of net joint moments calculated by the model for CMJ and SJC on the basis of joint angle and NSREMG time histories. CMJ: countermovement jump; SJC: squat jump starting from a posture that was identical to that at the start of push-off in CMJ. The vertical bars indicate the start of push-off (start of upward movement of the mass center of the body).

start of push-off between CMJ and SJC could be explained by a difference between CMJ and SJC in the time available to develop muscle force. To answer this question NSREMG time histories of the preparatory phase in CMJ were replaced by those in SJC. The resulting calculated joint moment time histories are labeled CMJm in Figure 5c (top). Moments at hip, knee, and ankle joints in the CMJm simulation were much lower at the start of push-off ($P < 0.05$) than in the CMJ simulation, and needed a considerable amount of time to catch up with the CMJ values. The effects on joint work, i.e., the area under the moment-angle diagrams shown in Figure 5c (bottom), were less dramatic. Work at the hip and knee joints decreased on average by 21 and 7 J, respectively ($P < 0.05$). The fact that joint work decreased more at the hip joints than at the knee joints may be explained as follows: because knee extension started to contribute later to upward movement of the center of mass than hip extension, the knee extensors had more time to develop force before the start of shortening (compare moment-time curves with moment-angle curves in Fig. 5c). The

reduction of joint moments was, of course, due to a reduction of active state of the muscles, which in turn was caused by the fact that NSREMG-levels in the preparatory phase were lower in SJC than in CMJ. The results of this simulation experiment suggest that the greater achievement in CMJ compared with SJC was mainly due to the fact that the countermovement allowed the extensor muscles to build up active state and force prior to shortening; in SJC shortening started as soon as the level of muscle stimulation was increased above that required for maintenance of the starting position, and consequently less force and thus less work was produced over the first part of the shortening distance.

The Role of Storage and Reutilization of Elastic Energy

During the countermovement in CMJ active muscles are pre-stretched and absorb energy, part of which is temporarily stored in series elastic elements and later reutilized in the phase where the muscles act concentrically. In the literature it is suggested that the elastic energy released during the concentric phase helps to increase the work produced in CMJ over that produced in SJ (2,25). First of all, it should be stressed that the amount of energy stored in series elastic elements at the start of the concentric phase is not determined by the amount of “negative work” performed but by the force at the start of push-off. As muscle forces are greater at the start of push-off in CMJ than in SJC, more elastic energy is stored in CMJ. In the model, the difference between CMJ and SJC in elastic energy at the start of push-off, added over all muscles, was 13 J. This means that almost all of the total amount of “negative work” in CMJ, on average 274 J, was converted to heat. The difference of 13 J was almost 50% of the difference in work corresponding to the observed difference in jump height between CMJ and SJC. The idea that the difference in elastic energy partially explained the difference in work produced is, however, wrong. This can easily be illustrated with the help of forward dynamic simulation of a single-joint action presented below.

For the simulation of a single-joint action we used a simplified version of the model, in which only the head-arms-trunk segment was free to rotate about the hip, and in which the hip joint moment was determined by the hip extensor muscles only (Fig. 2b). STIM was increased to a maximum of 0.56 (i.e., $f \cdot 1.0$ with $f = 0.56$, the average value for the hip extensors shown in Table 3) according to the following relationship: $STIM(t) = a_0 + a_1 \cdot \sin^2(\alpha t)$ for $0 < t < (0.5 \cdot \pi / \alpha)$ and $STIM(t) = a_0 + a_1$ for $t \geq 0.5 \cdot \pi / \alpha$, where a_0 is the STIM-level resulting in a joint moment that exactly balanced the moment due to gravity, $a_1 = (f \cdot 1.0 - A_0)$, and α is a constant equal to 5, a value that caused the rate of increase of STIM to approximate the observed rate of increase of SREMG-levels of the hip

TABLE 3. Mean values ($\pm$ SD) for the group of subjects ($N = 6$) of maximal isometric forces $F_{\max}$, and multiplication factors f , for NSREMG, of the six muscle-tendon complexes of the lower extremity incorporated in the model (GLutei, HAMstrings, VASti, RECTus femoris, SOLeus, and GAStrocnemius).

	SOL	GAS	VAS	REC	GLU	HAM
f	0.67 ± 0.19	0.50 ± 0.05	0.49 ± 0.04	0.46 ± 0.11	0.54 ± 0.09	0.58 ± 0.17
$F_{\max}$ [N]	5991 ± 1820	2996 ± 910	10490 ± 2844	3469 ± 948	5532 ± 1236	4426 ± 1010

The values were found by the optimization of model parameters for each individual subject, using as criterion the time-integral of squared differences between calculated and measured net joint moments. $F_{\max}$ values of homologous muscle-tendon complexes of the two legs have been added. For further details, see text.

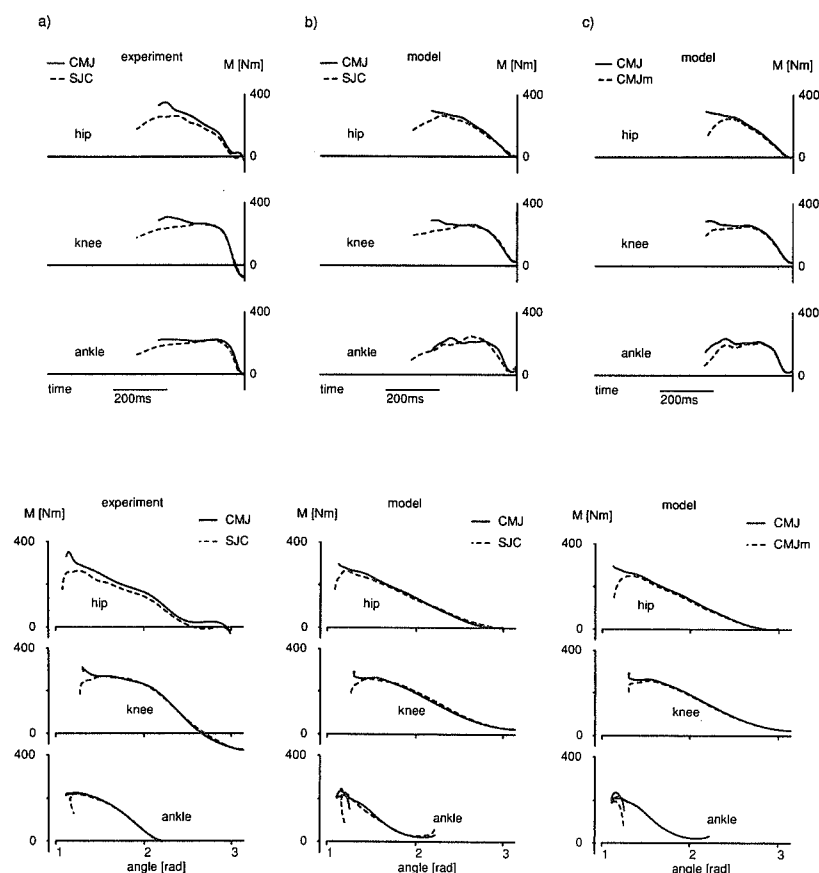
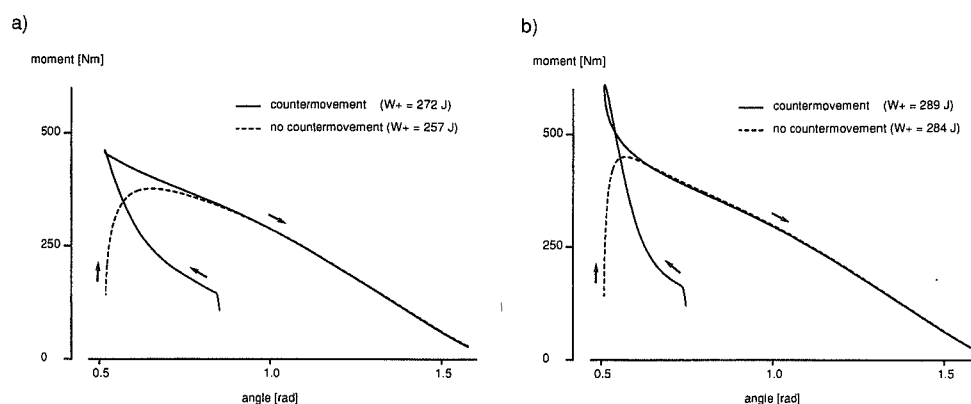


Figure 5—Average joint moment-time curves (top) and joint moment-angle curves (bottom) for the push-off phase in CMJ and SJC. (a) Experimental values for the group of subjects ($N = 6$). (b) Moments calculated by the model on the basis of individual joint angle and NSREMG time histories. (c) Moments calculated by the model when the NSREMG-time history of the preparatory phase in CMJ is replaced by that in SJC (CMJm). Model calculations were carried out for the individual subjects and then averaged.

extensor muscles. In a “countermovement” condition, the head-arms-trunk segment was given an initial angular velocity of $-3 \text{ rad}\cdot\text{s}^{-1}$ (clockwise) at $t = 0$. The increase in moment developed by the hip extensors caused the angular acceleration to be positive (anticlockwise), so that the angular velocity was reduced to zero at a minimum angle $\theta_{\min}$, and subsequently became positive. The moment-angle curve obtained in this condition is plotted in Figure 6a. Also plotted is the curve labeled “no countermovement,” obtained when the contraction was started with zero initial angular velocity at $\theta_{\min}$. In the “no countermovement” condition, less work was produced because it took the moment a certain angular delay to catch up with its “countermovement” companion, just as in the experimental results (Fig. 5b, bottom). Because of the difference in moment at the start of the concentric phase, there was also a difference in the amount of energy stored in series elastic elements between the two conditions.

The next step was to double the speed of the stimulation dynamics (by making α twice as large) as well as the excitation dynamics (by making m twice as large), while for simplicity ensuring that $\theta_{\min}$ remained the same. In that case, we obtained the results depicted in Figure 6b. This led to an increase of the difference between the conditions in the amount of energy stored, as evidenced by the increase of the difference in joint moment at the start of the concentric phase, but to a decrease of the difference in work produced. Obviously, it was the dynamics of force development that determined the differences in the amount of work produced, not the storage and reutilization of elastic energy! If the concentric angular displacement is the same, an increase in the amount of elastic energy stored at the start of the concentric phase merely reduces the amount of energy to be produced by the contractile elements, because lengthening of the series elastic elements occurs at the expense of the length over which the contractile elements can do work. Thus,

Figure 6—Moment-angle curves obtained in a forward dynamic simulation, using the simplified version of the model shown in Figure 2b. The increase of stimulation of the hip extensors above that required for equilibrium started at $t = 0$ (start of each curve). In the “countermovement” condition, the head-arms-trunk segment was given an initial angular velocity of $-3 \text{ rad}\cdot\text{s}^{-1}$ at $t = 0$, in the “no countermovement” condition the initial angular velocity was zero. (a) Stimulation dynamics correspond approximately to dynamics of SREMG. (b) Stimulation and excitation dynamics boosted by a factor of 2.



stored elastic energy increases the efficiency of doing positive work, but not the total amount of positive work that can be produced. These conclusions about the role of reutilization of elastic energy are in line with conclusions reached by Avis et al. (3) for isolated leg press actions and by Anderson and Pandy (1) for jumping.

The Role of Reflexes in Response to Pre-Stretch

In the literature, it is suggested that spinal reflexes or longer-latency responses to stretch of active muscles may help to increase the level of muscle stimulation over and above the level achieved in a contraction without stretch. The question whether or not this effect contributes to the difference in performance between CMJ and SJ is hard to answer, because it is not clear over which part of the jump such responses should be operative. The average NSREMG-level during push-off was not significantly higher in CMJ than in SJ in any of the muscles ($P > 0.05$). However, it cannot be ruled out that reflexes help to increase muscle stimulation during the countermovement phase in CMJ, and thus partly contribute to the development of a high level of active state and muscle force before the start of push-off.

The Role of Potentiation of the Contractile Machinery by Pre-Stretch

The overall enhancement of muscle moments at the start of push-off in CMJ compared with SJ was more than 60%, which is of the same order of magnitude of the force enhancement attained in studies of potentiation of the contractile machinery in isolated muscles (8–10,18) and single muscle fibers (14,15). The question may be raised whether the same mechanism is responsible for the enhancement. In our opinion, this is not the case. In the experiments on isolated muscles or muscle fibers the speed of pre-stretch is high and the pre-stretch is immediately followed by a concentric phase. In the present study, on the other hand, the speed of pre-stretch was relatively low, and a relatively long time delay (more than 200 ms) occurred between the end of pre-stretch and

the instant of maximal power production at the joints, which will tend to reduce the effect of pre-stretch on potentiation of the contractile machinery. In any case, in the simulation study the effect of pre-stretch on the concentric phase was due solely to the fact that the eccentric contraction leads to a greater force at the start of the concentric phase; there was no need to give the contractile elements a “memory” to explain the enhancement of joint moments, work, and jump height in CMJ compared with SJ.

Concluding Remarks

The experimental results obtained in this study confirm that a greater jump height can be achieved in CMJ than in CSJ, and indicate that the greater achievement in CMJ is attributed to the fact that individual muscles are able to produce more work after the countermovement, rather than to a more profitable body configuration at the start of push-off or better muscle coordination during push-off. The enhancement of joint work was mainly attributed to the fact that joint moments were higher during the first part of joint extension in CMJ than in CSJ. A recent attempt by Anderson and Pandy (1) to explain the greater jump height in CMJ using a forward dynamic model was unsuccessful; their model was actually able to jump higher in CSJ than in CMJ. However, their optimal control problem was a highly complex one, and in our opinion it cannot be excluded that the optimal control solution found for CMJ was not a global optimal solution. In the present study, a different type of modeling approach was selected to test possible explanations for the enhancement of muscle work after a countermovement. Using as input for a model the kinematics and electromyograms of subjects recorded during CMJ and CSJ, the experimental net joint moment time histories in CMJ and CSJ were approximated, and it was shown that the calculated moment-angle curves of CMJ could effectively be changed into those of CSJ by replacing the NSREMG time history during the preparatory phase in CMJ by that

in CSJ (the reverse was also true, but these results were not presented). Storage and reutilization of elastic energy could be ruled out as explanation for the enhancement of performance in CMJ over that in SJ. The crucial contribution of the countermovement seemed to be that it allowed the muscles to build up a high level of active state and force before the start of shortening, so that they

were able to produce more work over the first part of their shortening distance.

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